

KAI CHEN

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PROFESSIONAL SUMMARY

Reliability Engineer with 3+ years of battery cell safety & reliability experience at CATL (*Contemporary Amperex Technology Co., Limited*). Proven track record of leading cross-functional teams to deliver award-winning quality improvements, including a 95% failure rate reduction in mining-truck cells and extending commercial-vehicle cell warranty from 8 years / 800,000 km to 15 years / 1.5 million km. Skilled in FTA, FMEA, 8D methodology, and accelerated life testing. Currently leading a 6-person team overseeing 40+ projects across long-life, high-capacity, and fast-charging technology roadmaps.

WORK EXPERIENCE

CATL (Contemporary Amperex Technology Co., Limited) — Reliability & Failure Analysis Department Ningde, China
Cell Reliability Team Lead, Commercial Vehicle Business Jan 2026 – Present

- Led a 6-person team managing cell-level safety & reliability for the entire commercial-vehicle portfolio across **long-life, high-capacity, and fast-charging** technology roadmaps — overseeing 40+ R&D, mass-production, and after-sales projects.
- Delivered **20-year warranty long-life cells, 672Ah high-capacity cells, and 8C fast-charging cells** to commercial-vehicle mass production, expanding CATL's technology frontier.
- Primary reliability interface for top customers (**Yutong, CRRC, Volvo**); established team knowledge base and mentored 3 junior engineers to project ownership.

Reliability Engineer, **"Tianxing" Long-Life Commercial Vehicle Cell Program** Sep 2023 – Dec 2025

- Led end-to-end reliability risk management for CATL's flagship commercial-vehicle cell across design, prototype validation, mass production, and after-sales.
- Designed reliability verification schemes and established design, process, and application boundary specs through quantitative metrics analysis and accelerated life testing.
- Extended warranty from **8 years / 800,000 km to 15 years / 1.5 million km** — received **CATL 2025 President's Gold Award for Technological Innovation** (company's highest technical honor).

Reliability Engineer, **Mining-Truck Cell Quality Breakthrough Project** Sep 2024 – Dec 2025

- Led mining-truck cell reliability investigation for field failures (capacity fade, anode lithium plating); used **multi-level FTA** to identify root cause: insufficient charging window under continuous high-frequency duty cycles.
- Designed and validated **10+ corrective actions** across design, electrolyte, charging strategy, and process — all proven effective via accelerated life testing.
- Rolled out improvements across R&D and production; reduced failure rate by **over 95%**. Awarded **CATL 2025 Quality Breakthrough Award** for major quality improvement.

EDUCATION

Trinity College Dublin, Ireland Sep 2022 – Nov 2023

M.Sc. in Structural Engineering, School of Engineering **GPA: Top 10% | E3 Postgraduate Scholarship (€5,000)**

China University of Geosciences, Beijing Sep 2018 – Jun 2022

B.Eng. in Civil Engineering, School of Engineering and Technology **GPA: Top 20% | Project 211 · Double First-Class University**

HONORS & AWARDS

CATL President's Gold Award for Technological Innovation (2025) — Company's highest technical honor | **CATL Quality Breakthrough Award** (2025) — Major quality improvement | **Trinity College Dublin E3 Scholarship** (2022)

SKILLS & CERTIFICATIONS

Reliability & Quality: FTA (Fault Tree Analysis), DFMEA / PFMEA, 8D Problem Solving, Seven QC Tools, Accelerated Life Testing, Root Cause Analysis (RCA), Design Validation, Closed-Loop Quality Improvement

Technical Domains: Lithium-ion Battery Cell Safety, Commercial Vehicle Batteries, Cell Failure Analysis, Electrolyte Formulation, Charging Strategy, Battery Lifespan Modeling, Mathematical Modeling, Data Analysis

Languages: Mandarin (Native), English (Professional Working Proficiency — Duolingo English Test: 110 / IELTS 7.0 equivalent)

Tools: AI-assisted engineering (Claude Code), Data analysis & statistical modeling tools, MS Office Suite